

Detecting Fraud and Anomalies in Not For Profit Organization Online Donation Transactions: A Machine Learning approach*

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Abstract—In recent years, the growing influx of online donations to not-for-profit organizations (NFPOs) has advanced social initiatives, community development, and philanthropic missions. However, the complexity and scale of these transactions also expose NFPOs to fraud and misuse. Limited transparency and the absence of automated detection tools hinder the identification of subtle anomalous patterns, enabling misappropriation of charitable funds and eroding donor trust. Traditional monitoring systems often overlook irregularities in large-scale donation datasets. To address this gap, this study proposes a hybrid detection framework that extends Aziz’s modular architecture through enhanced preprocessing and dual-layer detection. In preprocessing, redundancy is addressed by integrating feature selection with mutual information, temporal scale detection, redundancy analysis, and correlation analysis over time, followed by class balancing when applied to real imbalanced datasets. At the detection stage, sequential anomalies are captured through a Hidden Markov Model (HMM), while collusive behaviors are detected using CoDetect’s relational network analysis. By combining redundancy-aware preprocessing with complementary temporal and relational models, the framework strengthens real-time identification of fraudulent activities while reducing false positives, thereby preserving financial integrity and donor confidence in NFPOs.

I. INTRODUCTION

Not-for-profit organizations (NFPOs) play a critical role in advancing social and philanthropic goals but remain persistently vulnerable to fraud despite preventive mechanisms. Traditional approaches such as audits, threshold-based monitoring, and layered databases often fail to detect nuanced anomalies including duplicate donations, oddly timed contributions, or irregular spending behaviors, especially in card-not-present transactions [1]–[3]. These blind spots expose NFPOs to financial misconduct that undermines credibility, erodes donor trust, and deters future contributions. Empirical studies further reveal that fraud-related losses range from thousands of dollars in local charities to millions in larger institutions, with average cases exceeding \$100,000 [1]–[3].

In parallel, financial domains such as credit card and online banking transactions face similar challenges, where class imbalance, adaptive adversarial strategies, and complex behavioral deviations hinder effective fraud detection. Advances in machine learning have demonstrated value in

these settings by capturing sequential dependencies [4] and relational patterns among entities [5]. However, most existing NFPO frameworks do not fully exploit these perspectives, nor do they systematically address redundancy and imbalance in high-dimensional datasets.

This study proposes a specialized fraud detection framework for NFPO donation transactions by extending Aziz’s modular architecture [6]. The contributions are threefold. First, the preprocessing stage integrates feature selection based on mutual information, temporal scale detection, redundancy analysis, and correlation analysis over time to remove duplicate and overlapping attributes before class balancing. Second, the detection layer employs a dual approach that combines an HMM for modeling irregular sequential donation patterns with CoDetect for detecting coordinated or collusive schemes. Third, a unified decision engine consolidates both outputs, enabling real-time classifications with improved accuracy and reduced false positives. This framework is designed to strengthen financial integrity, preserve donor trust, and provide a scalable baseline for NFPOs facing increasingly complex fraud risks.

A. Research Objectives and Questions

The objective of this study is to design and evaluate a fraud detection framework tailored for NFPO donation transactions that extends existing modular architectures with redundancy-aware preprocessing and dual detection mechanisms. Specifically, the framework aims to (1) improve feature quality by eliminating redundancy and preserving temporal information, (2) enhance anomaly detection through complementary sequential and relational models, and (3) provide a scalable and interpretable architecture suitable for NFPO contexts.

Based on these objectives, the study addresses the following research questions:

- **RQ1:** How can redundancy handling through mutual information, temporal scale detection, redundancy analysis, and correlation analysis over time improve feature quality for NFPO fraud detection?
- **RQ2:** How can temporal donation anomalies be effectively captured using sequential models such as Hidden Markov Models (HMM)?

- **RQ3:** How can collusive or relational fraud be detected by leveraging relational network analysis techniques such as CoDetect?
- **RQ4:** How can a unified decision engine balance accuracy against false positives in highly imbalanced NFPO donation datasets?

These research questions guide the design of the proposed methodology and frame the evaluation criteria for the framework.

II. RELATED WORK

Early studies in fraud detection applied machine learning to financial domains such as credit card transactions. For example, [7] demonstrated that neural networks could detect diverse fraud types, including lost, stolen, counterfeit, and application fraud. Building on this foundation, the *Hidden Markov Model (HMM)* was introduced for fraud detection, capturing sequential dependencies in transaction data and effectively identifying deviations from normal behavioral patterns [4], [8]. These probabilistic models are particularly well suited to domains where transaction order and timing contain valuable signals of anomalous activity.

Online financial fraud research, particularly in credit card and banking, further established key methodological challenges. These include severe class imbalance, adaptive adversarial strategies, and risks in card-not-present transactions. Studies emphasized that addressing imbalance and behavioral deviations is essential to improving fraud detection performance [9]–[11]. Preprocessing and feature engineering therefore play a critical role in ensuring data quality, with approaches such as SMOTE [12] widely adopted to mitigate imbalance. Surveys of fraud detection methods highlight that class imbalance, concept drift, and high false positives remain enduring challenges in real-world applications [13], [14].

In addition to balancing techniques, feature selection and temporal analysis have been emphasized as complementary preprocessing strategies. Methods that Handle redundancy with mutual information, temporal scale detection, redundancy analysis, and correlation analysis over time are particularly relevant. Information-theoretic approaches such as minimum-redundancy maximum-relevance (mRMR) [15] and normalized mutual information feature selection (NMIFS) [16], reviewed comprehensively by Vergara and Estévez [17], ensure compact and non-redundant feature subsets. Beyond static redundancy handling, redundancy analysis techniques such as those presented by Baho et al. [18] provide tools to identify distinct temporal patterns and scales, while Brown et al. [19] highlight how correlation analysis across time series can uncover hidden temporal relationships. Similarly, Langii [20] demonstrates how redundancy quantification and reduction in multivariate data improves machine learning robustness under incomplete or overlapping signals. Collectively, these works motivate integrating independent feature selection and temporal-scale analysis into preprocessing so that fraud detection models operate on compact, informative, and temporally meaningful feature sets.

Beyond preprocessing, research has shifted toward more advanced detection paradigms. Supervised models such as logistic regression and support vector machines achieved improved accuracy but depend heavily on labeled data, making them less adaptable in dynamic fraud contexts [21]. In contrast, unsupervised and hybrid methods such as clustering, anomaly detection, and neural networks demonstrated stronger generalization for detecting unseen or subtle fraud patterns [22], [23]. Hybrid frameworks in particular have gained traction for combining complementary strengths of different models.

One such example is CoDetect, which integrates two core components: (1) anomaly feature detection to identify suspicious deviations at the transaction level, and (2) relational network analysis to uncover hidden associations among accounts, devices, or shared attributes [5]. This dual perspective allows CoDetect to detect collusive and coordinated fraud schemes that traditional transaction-only models fail to capture. However, CoDetect does not explicitly model temporal sequence patterns, such as irregular donation timing or repeated behavioral anomalies. This limitation highlights the complementary value of temporal models like HMMs, which provide strong individual profiling but lack relational analysis.

In terms of system-level design, [6] proposed a modular and scalable architecture for credit card fraud detection. Their framework consists of a preprocessing pipeline, a detection layer with multiple models, and a decision engine for real-time classification, as illustrated in Figure 1. While effective as a general-purpose baseline, the architecture does not directly address sector-specific challenges such as NFPO fraud, where both sequential donation behaviors and collusive schemes play critical roles.

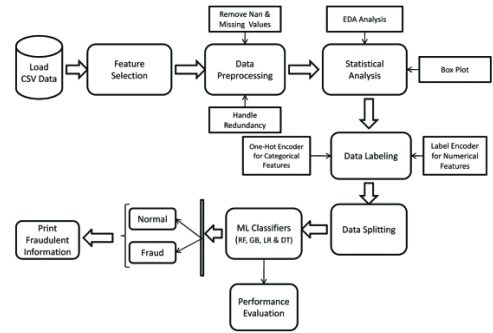


Fig. 1. System Architecture as proposed by [6].

Overall, prior research demonstrates three important trends: (1) neural networks and probabilistic models established early success in fraud detection, (2) hybrid and relational approaches such as CoDetect expanded detection to collusive fraud, and (3) modular architectures such as Aziz’s system provide scalability but lack specialized modeling of temporal and relational perspectives. In addition, limited attention has been

given to redundancy handling and temporal-scale analysis, even though methods such as mutual information, redundancy analysis, and correlation analysis over time have proven effective in improving data quality. This contrast highlights a clear research gap. Effective fraud detection frameworks must simultaneously capture **sequential behavioral anomalies**, **network-based collusion**, and **Handle redundancy with mutual information, temporal scale detection, redundancy analysis, correlation analysis over time** while maintaining scalability. The present study addresses this gap by extending Aziz’s architecture with a dual-layered detection approach that combines HMM and CoDetect for enhanced NFPO fraud detection.

III. METHODOLOGY

Building on insights from prior studies, this research proposes a fraud and anomaly detection framework tailored for NFPOs that extends the modular architecture of [6] through advanced preprocessing, statistical analysis, and complementary detection models, as shown in Figure 2.

The preprocessing stage systematically addresses missing values and redundancies. To explicitly handle redundancy with mutual information, temporal scale detection, redundancy analysis, and correlation analysis over time, we move independent feature selection and temporal correlation checks into this stage. Redundancy handling begins with information-theoretic measures: mRMR [15] and NMIFS [16] evaluate the trade-off between a feature’s predictive relevance to the class and its redundancy with already-selected features [17]. This ensures that attributes providing duplicate or overlapping information are removed early, preserving only compact and informative subsets. Beyond mutual information filtering, redundancy analysis techniques [18] are employed to assess the contribution of features across different temporal scales, enabling us to distinguish between short-term fluctuations (e.g., irregular bursts of donations) and long-term seasonal trends. Features that fail to add explanatory value at any scale are discarded. In addition, correlation analysis over time [19] is used to capture dependencies between time-dependent variables, helping to detect features that evolve in highly similar ways and thus do not contribute unique information. Finally, redundancy quantification methods, such as those demonstrated by Langii [20], guide the reduction of overlapping signals in multivariate data streams, improving the robustness of downstream models under incomplete or overlapping inputs. Together, these steps ensure that the retained features are compact, non-redundant, and temporally meaningful before the data balancing stage. After redundancy has been addressed, the Synthetic Minority Oversampling Technique (SMOTE) [12] is applied to mitigate severe class imbalance, ensuring that both authentic and synthetic patterns contribute to the training process.

It is important to note that the role of class balancing differs depending on the dataset used. In real-world NFPO donation datasets, fraudulent transactions are expected to represent only a very small fraction of all records, resulting in a severe class imbalance that can bias model training toward the majority

(legitimate) class. In such cases, oversampling strategies such as SMOTE [12] are necessary to generate synthetic minority examples and ensure that fraud-related patterns are sufficiently represented during learning. However, in the present study, publicly available NFPO fraud datasets are not accessible, and synthetic data are used for framework development and evaluation. Since the class distribution in synthetic data can be controlled at generation time, additional balancing via SMOTE is not required in this context. This distinction highlights that SMOTE will be critical when applying the proposed framework to real-world NFPO donation data, whereas it is optional when working with controlled synthetic datasets.

At the detection stage, the framework incorporates both sequential and relational learning paradigms. An HMM [4] is employed to capture temporal dependencies in donation sequences, which are often irregular but sequential in nature. In parallel, CoDetect [5] integrates anomaly feature detection with relational network analysis, enabling identification of collusive behaviors such as shared devices, repeated account registrations, or overlapping transaction attributes.

Finally, a decision engine consolidates outputs from both HMM and CoDetect, producing real-time classifications that distinguish between fraudulent and normal transactions. This dual-layered detection approach enhances robustness by detecting both individual anomalies and coordinated fraud schemes. A performance evaluation stage quantifies detection accuracy, while a reporting mechanism presents structured insights on anomalous activities.

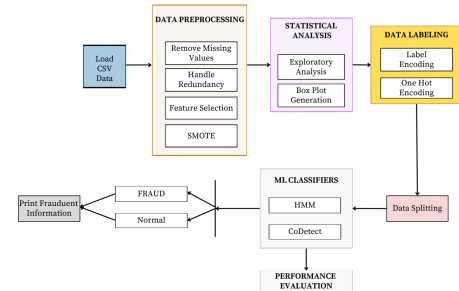


Fig. 2. Improvement for [6]’s System Architecture. *Note: SMOTE is applied only when training on real-world NFPO datasets, where class imbalance is severe. For synthetic datasets with controlled class distributions, additional balancing is not required.*

By enriching Aziz’s modular architecture with advanced preprocessing, explicit redundancy handling with mutual information, temporal scale detection, redundancy analysis, correlation analysis over time, temporal sequence modeling, and relational anomaly detection, the proposed framework achieves improved adaptability to imbalanced NFPO datasets and heightened sensitivity to coordinated fraud patterns.

IV. CONCLUSION

This study presented a conceptual framework for fraud and anomaly detection in NFPO donation transactions by extending the modular pipeline of [6]. The framework strengthens

preprocessing through the integration of feature selection with mutual information, temporal scale detection, redundancy analysis, and correlation analysis over time, ensuring that retained features are compact and temporally meaningful. When applied to real datasets, oversampling techniques such as SMOTE [12] can further mitigate severe class imbalance, while synthetic datasets with controlled class distributions do not require additional balancing. At the detection layer, sequential anomalies are captured through HMM [4], and collusive behaviors are detected using CoDetect [5], with both perspectives unified in a decision engine for real-time fraud classification.

As this work is theoretical in nature, future development should refine and validate the proposed preprocessing and detection stages on representative NFPO datasets. Further directions include extending redundancy handling with alternative estimators, exploring additional anomaly detection techniques, and evaluating robustness under varying imbalance conditions. Ultimately, by integrating redundancy-aware preprocessing with temporal and relational detection models, the proposed framework provides a pathway toward scalable, reliable, and domain-specific fraud detection for NFPOs.

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